An improved predictive model for assessing the impact of deforestation and CO_2 emissions on flood hazards in Pakistan

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ABSTRACT: Climate change, deforestation, and rising CO_2 emissions have significantly increased the frequency and severity of floods in Pakistan. Traditional forecasting models often fail to account for critical environmental variables such as land-use changes, forest degradation, and atmospheric carbon concentrations. This study introduces a novel multi-dimensional machine learning (ML) framework that systematically integrates environmental degradation indicators, deforestation rates, CO_2 emissions, and land use changes, with traditional hydrological and meteorological variables for flood prediction in Pakistan. Addressing a critical methodological gap in regional flood forecasting, it presents the first comprehensive integration of these factors using advanced ensemble ML techniques. A diverse dataset comprising 5,500 records from 2010–2023, sourced from national and international repositories, was compiled and preprocessed through median imputation, StandardScaler normalization, feature selection, and SMOTE balancing. The proposed hybrid boosted ML model and stacking-based ensemble framework demonstrate significant technical innovation and impact, achieving 95.73% accuracy, F1-scores exceeding 0.97, ROC-AUC of 0.982, and PR-AUC of 0.98. Through systematic evaluation of twelve algorithms, this framework establishes new benchmarking standards in flood prediction research. Feature importance analysis reveals CO_2 emissions (correlation: 0.79), rainfall (0.78), and deforestation rates (0.74) as dominant predictors, quantitatively confirming the critical influence of environmental degradation on flood hazards—a key insight previously unaddressed in regional studies. This research advances both scientific understanding and practical disaster management by enhancing model interpretability and delivering actionable insights for evidence-based early warning systems and policy recommendations. It directly addresses challenges such as data imbalance, feature uncertainty, and multi-source integration, offering a robust and scalable solution for flood risk prediction. The methodology further provides a foundation for future work, including geographic expansion, real-time environmental data incorporation, advanced ensemble learning, and mobile application development, contributing toward climate-resilient disaster management at both national and global scales..

Keywords: Flood Prediction; Machine Learning; Climate Change; Deforestation; CO_2 Emissions; Rainfall; Environmental Indicators; Stacking Classifier; Early Warning Systems; Flood Risk Management; Disaster Preparedness.

1. INTRODUCTION

Floods can be very destructive anywhere in the world, leading to big economic setbacks, harm to the environment, and the loss of many lives in vulnerable communities. Pakistan, with its diverse geographical features, seasonal monsoon patterns, and increasing anthropogenic environmental pressures, faces critical flood vulnerability that has intensified dramatically over recent decades. Because of climate change and a quick increase in deforestation and CO_2 emissions, nations' hydrological landscapes are being greatly impacted, creating unprecedented challenges for traditional flood prediction and management systems [1], [2].

Fig. 1: Global flood risk distribution, highlighting Pakistan's high vulnerability. Source: World Bank [3].

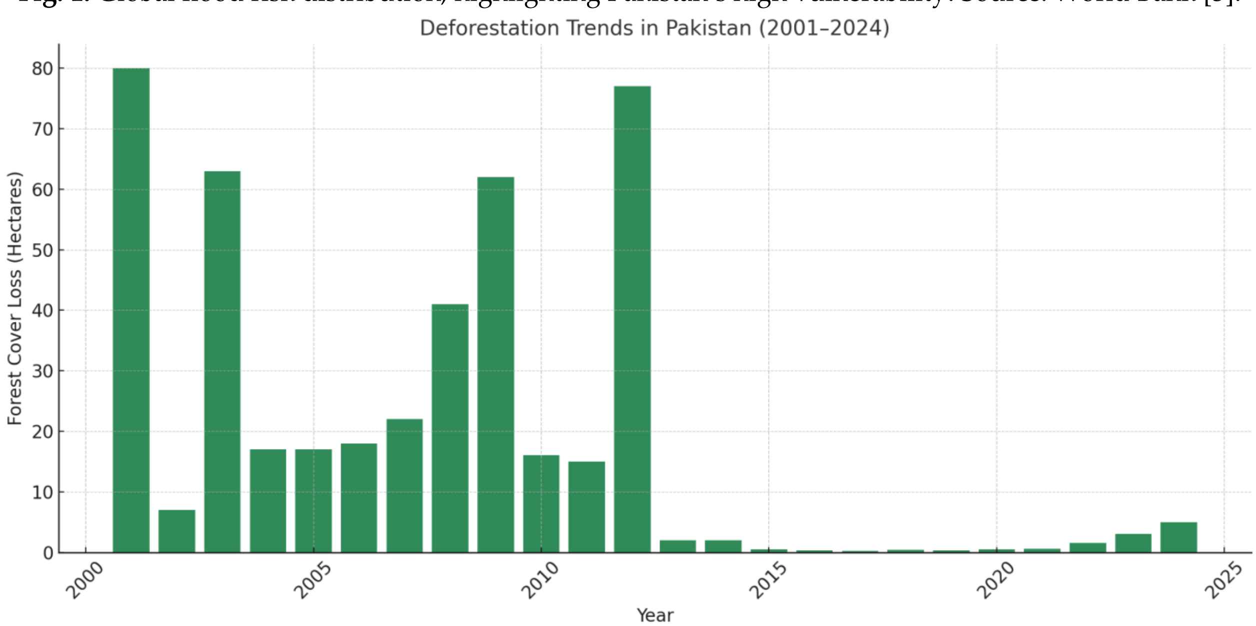


Fig. 2: Deforestation trends in Pakistan (2001–2024) showing forest cover loss (in hectares) over time. Source: Global Forest Watch (2024) [4].

Fig. 1 visualization ranks countries by population exposure to flood hazards, with South Asia emerging as a high-risk zone. Pakistan's position reflects its unique combination of climatic, geographic, and anthropogenic vulnerabilities. **Fig. 2** illustrates the accelerating deforestation in Punjab, Pakistan, with a notable spike in forest cover loss post-2010. The trend correlates with increased agricultural expansion and urbanization, exacerbating flood risks due to reduced natural water absorption capacity. The major 2010 floods and the 2022 floods are good examples of how serious the flood crisis has become in Pakistan, as they each displaced huge numbers of people and resulted in billions in damages. Current flood prediction systems in Pakistan demonstrate significant limitations in addressing multifaceted environmental challenges. Most traditional hydrologic models are concerned with how much it rains and how the rivers behave, but leave out the key environmental changes triggered by deforestation, carbon emissions, and changes in land use. These conventional approaches proved inadequate during recent major flood events, failing to accurately predict the timing, intensity, and spatial distribution of flooding, resulting in inadequate early warning systems and ineffective disaster preparedness [5], [6].

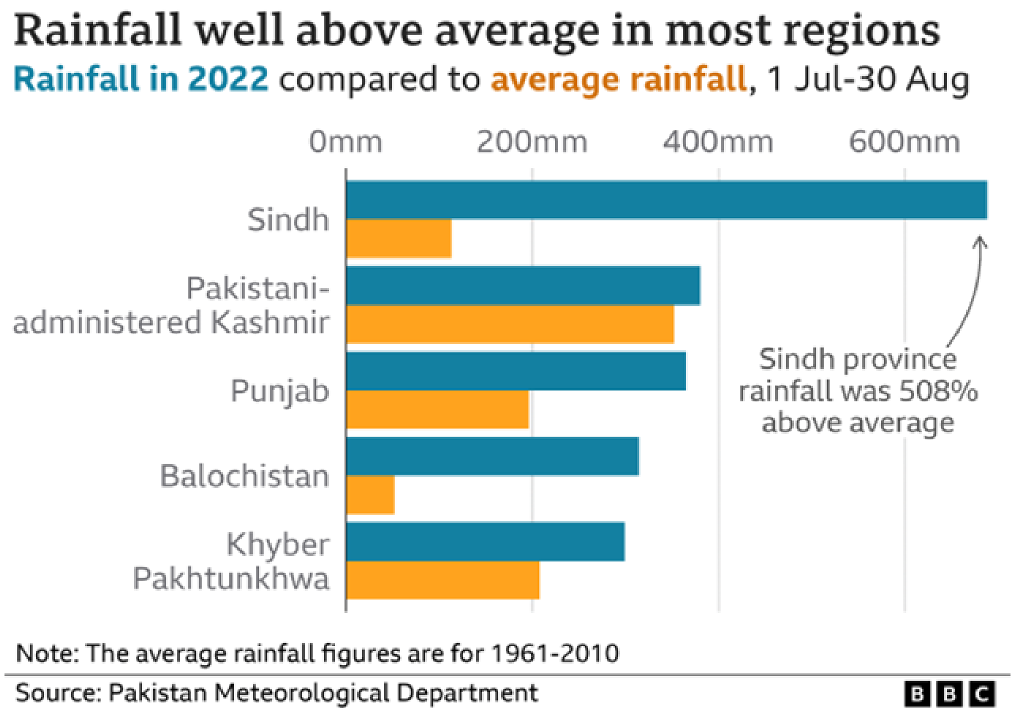


Fig. 3: Regional rainfall anomalies (July–August 2022) compared to 1961–2010 averages across Pakistan. Source: Pakistan Meteorological Department (2022) [7].

Fig. 3 highlights the extreme rainfall disparities in 2022, particularly in deforested regions like Punjab and Sindh. The spatial correlation between precipitation peaks and areas of significant forest cover loss (Figure 1) suggests a compounded flood risk due to diminished land absorption capacity. Despite the critical importance of flood prediction in Pakistan, existing literature reveals several significant gaps. First, traditional forecasting approaches primarily rely on meteorological data while ignoring the substantial impact of environmental degradation factors such as deforestation and carbon emissions on flood occurrence. Second, there is a lack of comprehensive comparative studies evaluating multiple machine learning algorithms specifically for Pakistan's unique environmental conditions. Third, current prediction models fail to integrate multi-dimensional environmental datasets that capture the complex interactions between climate change, land use changes, and flood risks. Finally, existing studies lack systematic feature importance analysis to identify the most critical environmental predictors for flood occurrence in the Pakistani context [8] [9].

The increasing frequency and intensity of flood disasters demonstrate that traditional forecasting approaches are inadequate for capturing the complex interactions between climate change, environmental degradation, and flood risk dynamics. Climate change has emerged as a primary driver amplifying flood risks through intensified glacier melt, destabilized weather systems, and unpredictable monsoon behaviors [10]. Simultaneously, deforestation and rapid urbanization have diminished natural water retention capacity and disrupted drainage patterns. The urgent need for sophisticated prediction methodologies that incorporate comprehensive environmental factors beyond conventional hydrological parameters motivates this research to develop more accurate and reliable flood forecasting systems.

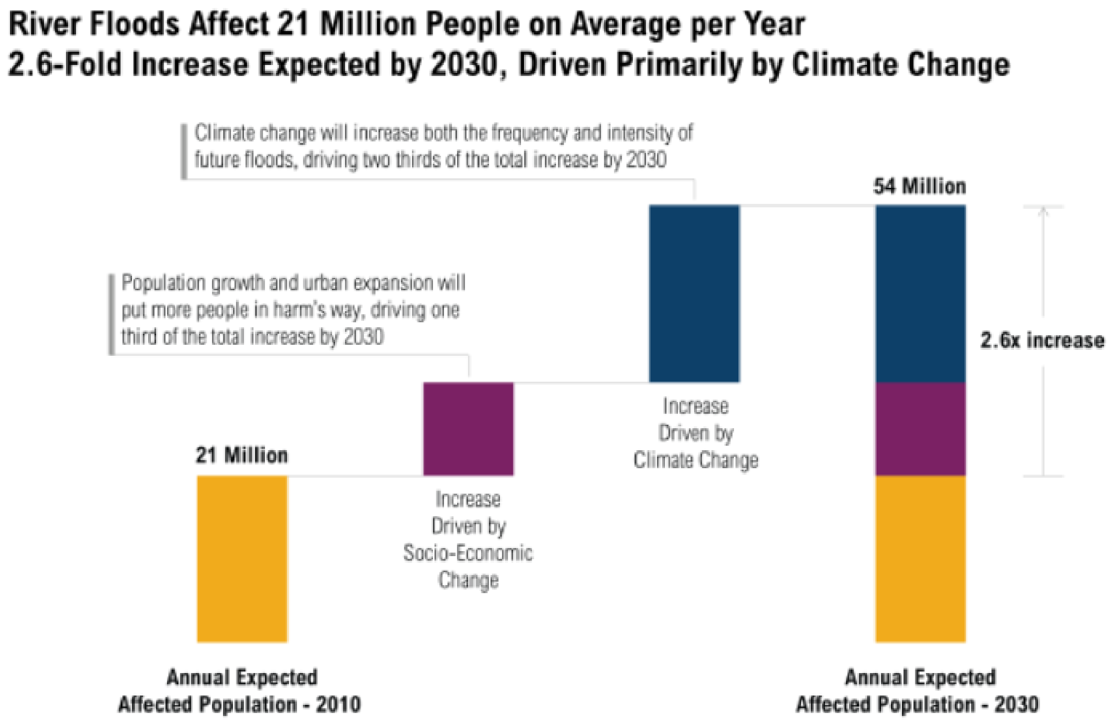


Fig. 4: Projected increase in flood-affected populations (2010–2030) driven by climate change and urbanization. Source: World Resource Institute [11].

Fig. 4 illustrates that this 2.6-fold surge highlights the compound threat of environmental change and population growth in vulnerable regions like Pakistan. The study aims to address these gaps by focusing on specific objectives. First, the research seeks to develop a comprehensive machine learning framework that integrates climate change indicators, deforestation data, CO_2 emission levels, and traditional hydrological variables for flood prediction in Pakistan. Second, the study systematically evaluates and compares multiple advanced machine learning algorithms to identify the most suitable models for Pakistan's environmental conditions. Third, the research analyzes the relative importance of environmental degradation factors in flood prediction accuracy through comprehensive feature importance analysis. Fourth, the study validates the superiority of the proposed multi-dimensional approach over traditional statistical methods through rigorous performance evaluation. Finally, the research aims to provide practical recommendations for disaster management authorities and policymakers based on the research findings.

- This study introduces a novel, multi-dimensional ML framework for flood prediction in Pakistan, being the first to systematically integrate environmental degradation indicators, deforestation rates, CO_2 emissions, and land use changes, with traditional hydrological variables. It establishes new benchmarking standards through a systematic comparative evaluation of twelve advanced algorithms, addressing a critical methodological gap in regional flood prediction research.
- The proposed hybrid boosted ML model efficiently assesses the impact of deforestation and CO_2 emissions on flood hazards, enhancing predictive performance. Moreover, the developed ensembled ML framework, utilizing the stacking of top-performing classifiers, ensures improved accuracy and robustness in flood hazard assessment.
- The research provides the first comprehensive feature importance analysis linking environmental factors, such as deforestation, CO_2 emissions, and land use changes, with flood occurrence in Pakistan, quantitatively

assessing the influence of individual climate change indicators on prediction accuracy. This advances both model interpretability and scientific understanding while offering actionable insights for disaster management authorities, supporting more effective early warning systems and evidence-based policy decisions for sustainable environmental management.

- The proposed framework establishes a robust foundation for future investigations, including expanding to other provinces, incorporating real-time environmental data, exploring advanced ensemble learning techniques, developing mobile applications for practical deployment, and adapting the methodology for disaster-prone regions globally. This contributes to advancing environmental risk assessment and climate-resilient disaster management strategies.

The rest of this paper is put together as follows: Section 2 explains the major findings from related studies about flood prediction and points out what is still missing in the field. Section 3 covers the research method used, such as how the data is collected, processed, and the implementation of the machine learning model. Section 4 covers the Results & discussion. In Section 5, the author discusses what the findings indicate, suggests implications, and offers possible solutions.

2. Related Work

In recent years, many researchers have proposed several methods for assessing the impact of deforestation and CO_2 emissions on flood hazards [12]. Combining remote sensing, ensemble machine learning, and hydrodynamic modeling with a view to assessing flood susceptibility in Pakistan, Ahmad et al. (2025) achieved the results. They show that a physically-based method has advantages when it is combined with AI-based methods in mapping spatial flood risk [13]. Sajjad et al. (2025) employ the use of remote sensing metrics such as MNDWI and NDVI to locate and determine the flash flood destructions in Dera Ismail Khan. Their results highlight the usefulness of satellite images to have the impact of the floods measured within a small time span [14]. Singh and Raghavan (2024) employ machine learning to analyze climate and land-use change in Tamil Nadu, India. Their research emphasizes predictive modeling for climate adaptation and focuses on how environmental changes exacerbate flood threats [1] . Nishat et al. (2024) study the effects of deforestation on local climate and weather patterns. According to the study, deforestation disrupts precipitation, raises surface temperatures, and alters air circulation, all of which lead to greater climate variability [2]. Muhammad et al. (2024) study how energy demand, poverty, and governance issues contribute to deforestation and carbon emissions. They show how ineffective policies and excessive resource use contribute to environmental degradation [5]. Ali et al. (2024) use statistical and machine learning models to identify flood-prone locations in Sindh, Pakistan, which improves risk assessment. However, the study lacks real-time data integration and extensive geographical validation [6]. Khan et al. (2024) created a machine learning-based flood risk prediction model for the Indus Basin, which will improve early warning systems. However, the study ignores socioeconomic issues, which might help improve disaster preparedness strategies [8].

Hina et al. (2024) used GIS and remote sensing to identify flood risk zones in Tehsil Shah Alam, Peshawar. While the findings are valuable for flood control, there are limitations in temporal resolution and ground-truth validation [9].

Bazai et al. (2024) look at the 2022 Swat River flood, focusing on monsoon-caused geological hazards and the socioeconomic effects. The research investigates fundamental disaster management learning points yet fails to include predictive models for estimating future flood risk [10].

Combining remote sensing, ensemble machine learning, and hydrodynamic modeling with a view to assessing flood susceptibility in Pakistan, Ahmad et al. (2025) achieved the results. They show that a physically-based method has advantages when it is combined with AI-based methods in mapping spatial flood risk [13]. The study by Mukhtar et al. (2024) uses the GIS and multi-criteria decision-making to determine the vulnerability of the Hunza Nagar region to floods. The paper combines climate information, terrain information and land-use data to form a compound model of flood risk [15]. Sajjad et al. (2024) analyse historical flood extent in Pakistan using spatial analysis and the modal of satellite imagery in MODIS. Their study gives a long term flood mapping information based on NDWI, especially in the Mianwali area [16]. In a GIS backdrop, Ahmad et al. (2024) apply a flood vulnerability assessment by using the

Analytical Hierarchy Process (AHP) in the Sindh watershed. They also prioritize environmental and infrastructural conditions to know the areas on high-risk [17].

To determine the susceptibility of landslides and floods in the north of Pakistan, Hafeez et al. (2024) use artificial neural networks and GIS methods. In their work, they have incorporated the data of topography and environment in order to enhance spatial hazard forecasts[18]. Ahsan and Javaid (2024) evaluate the Pakistan floods of 2022 in terms of the space-based indicators, land use, and soil moisture imagery. Their interests are placed on gauaging agricultural and urban damages by means of geospatial approaches [19]. This proposal by Gupta et al. (2024) is entitled FloodCast: A physics-guided machine learning framework of large- scale flood forecasting. The model combined satellite-derived data and principles of hydrology, and their comparison is performed with the use of data on disasters in Pakistan in 2022 [20].

The researchers of Rehman et al. (2023) studied the relationship between rainfall patterns and land use practices on flood risk. The study provides valuable information about urban development without integrating hydrological and meteorological models, which leads to reduced accuracy of their predictions [21]. Through a semi-probabilistic approach, Nossent (2023) conducts flood risk evaluations that include climate change ambiguities. The author demonstrates the necessity of probabilistic modeling to enhance flood risk prediction and management systems [22].

Nanditha et al. (2023) examine the origin factors as well as the consequences of the Pakistan flood event during August 2022. The research identifies heavy monsoon rains together with deforestation and inadequate infrastructure as the main factors that caused the disaster, while emphasizing the necessity of climate resilience measures [23]. According to Rehman and Qadir (2023), Pakistan faces deforestation problems, as they present a case for implementing sustainable forest practices. The authors stress that afforestation programs as well as legislative changes represent key solutions for climate change mitigation and environmental conservation [24]. The Government of Pakistan & GIZ (2023) explain Pakistan's climate vulnerability in a policy paper. The statement demonstrates how essential it is to establish national and international collaborations in both climate change adaptation strategies and disaster readiness programs [25].

The carbon emissions alongside forest cover alterations in Pakistan's dry temperate regions are studied by Ahmad et al. (2022). The study shows major trends of forest depletion while recommending sustainable forest practices to reduce carbon emissions [26]. Manzoor et al. (2022) examined how flood management strategies in Pakistan affect the social and economic aspects of the country. The authors make the case for implementing flood mitigation approaches that will enhance resilience and minimize damage incidents [27].

The research by Narisma and Wang (2022) demonstrates worldwide evidence that deforestation causes elevated flood risks primarily in underdeveloped countries. The authors emphasize the importance of creating sustainable land-use plans to prevent climate-related disasters [28]. Khan et al. (2021) evaluate the impact of climate change on flood management in Pakistan. They recommend adaptive mitigation strategies and stronger land use policies [29].

The literature studies highlight the complex relationship between deforestation, climate change, and flood risks. Deforestation significantly modifies local climatic conditions, disrupts hydrological cycles, and increases the frequency and severity of extreme weather events. The flooding disaster in Pakistan in August 2022 serves as an appropriate example of these consequences, underlining the crucial importance of implementing sustainable land-use regulations, massive afforestation programs, and the advancement of climate modeling tools. Furthermore, to effectively manage flood risks and build long-term environmental resilience, governance institutions must be reinforced, community involvement promoted, and predictive analytics and technological advancements used.

The following Table 1 summarizes the twenty references. Key aspects include years, titles, authors, suggested work, references, limits, technology, and gaps.

Table 1. Summary of Different Reference Papers

Sr. No	Year	Author	Ref.	Method	Technology	Limitations
1	2025	Ahmad, I., Farooq, R., Ashraf, M., et al.	[13]	Combines ML with dimensions of hydrodynamic modeling and remote sensing and incorporates ensemble ML in Pakistan to rate flood susceptibility.	HEC-RAS, Remote Sensing, EMs ML	Restricted scalability based on the need of the computational re-sources.
2	2025	Sajjad, A., Ahmad, M., Aslam, R. W., et al.	[14]	Satellite indices of flash flood mapping und damage assessment.	Remote Sensing (MNDWI, NDVI), Google Earth	The precision relies mainly on quality of satellite images and cannot be ground checked.
3	2024	Singh, R., & Raghavan, S.	[1]	Using machine learning to evaluate flood vulnerability.	Machine Learning, Flood risk analysis	Insufficient generalizability to other geographical areas.
4	2024	Nishat, P., Kumari, A., & Kumar, S.	[2]	Investigating how deforestation affects local weather patterns.	Meteorological data, Statistical models	The relevance and potential for expansion within specific regions.
5	2024	Muhammad, T., Umer, K., Alvi, S., & Wilson, C. J.	[5]	Investigate the causes of deforestation and carbon emissions.	Environmental assessment, Regression analysis.	Insufficient longitudinal data regarding the effects of climate change.
6	2024	M. Ali, S. M. H. Zaidi, F. Nawaz, and F. Sheikh	[6]	Uses statistical and machine learning techniques to predict flood-affected regions in Sindh.	Statistical models, Machine Learning algorithms	Requires real-time flood monitoring and validation in several regions
7	2024	M. Khan, A. U. Khan, B. Ullah, and S. Khan	[8]	A machine learning algorithm for predicting flood danger in the Indus basin	Machine Learning	Climate change and hydrological models need to be included.

8	2024	H. Hina, S. Gul, Z. Ali, and U. Ullah	[9]	Assessing flood risk using GIS and remote sensing	GIS, Remote Sensing	Requires better resolution data and confirmation using ground truth information.
9	2024	N. A. Bazai et al.	[10]	Research on geological hazards induced by monsoon events.	Geological analysis, historical flood data	Predictive modeling is needed to determine future flood risk.
10	2024	Mukhtar, M. A., Shangguan, D., Ding, Y., et al.	[15]	GIS-based MCDM-based integrated flood risk analysis.	GIS, AHP, Big Climate Data	It may fail to take into consideration the changing patterns of rainfall or changes that occur in land-use.
11	2024	Sajjad, A. et al.	[16]	Spatial indicators/MODIS based historical mapping of floods.	MODIS, NDWI, GIS	It just addresses the past information; it does not include the prediction of the flood in the future.
12	2024	Ahmad, A. et al.	[17]	Sindh watershed: multi-criteria vulnerability of floods.	AHP, GIS	Low temporal flexibility; concerned with the fixed vulnerability indicators.
13	2024	Hafeez, H., Gul, S., Ali, Z., & Ullah, U.	[18]	AML and GIS applied landslide/flood hazard mapping of Northern Pakistan.	ANN, GIS, Remote Sensing	Limited generalization may be experienced because of local training datasets. Datasets.
14	2024	Ahsan, M., & Javaid, A.	[19]	Assesses the effects of floods in 2022 based on multi-sensor topographic and satellite data.	Landsat-9, ESA-LULC, SMAP	Not designed to incorporate predictive modeling, only damages mapping.
15	2024	Gupta, S., et al.	[20]	Introduces Flood Cast-artificial intelligence-based framework synthesizing hydrological physics and satellite application.	GeoPINS, Physics- Satellite imagery, Informed Neural Networks.	Experimental; yet to be used in the developing world e.g. Pakistan.

16	2023	N. Rehman, U. Zada, K. Haleem	[21]	Examines how land development and rainfall affect flood dangers.	Statistical Analysis, Land Use Mapping	To provide more accurate projections, hydrological and climatic models must be integrated.
17	2023	Nossent, J.	[22]	Flood risk analysis with climate uncertainty	Risk Analysis, Probabilistic modeling	Real-time forecasting lacks practical applicability.
18	2023	Nanditha, J., Prakash, A., Kushwaha, I., et al.	[23]	Investigate the reasons for the 2022 Pakistan floods.	Data analysis, Historical records	Need for proactive flood management measures.
19	2023	Rehman, A., & Qadir, S.	[24]	Review of the Need for Transformational Forest Management in Pakistan	Review, Environmental Policy	Insufficient activity on forest management.
20	2023	Government of Pakistan & GIZ	[25]	Gives an overview of Pakistan's susceptibility to climate change, including flood risks.	Climate policy report	There are no statistical or machine learning-based flood risk predictions.
21	2022	Ahmad, A., et al.	[26]	Analysis of carbon emissions from deforestation	Environmental modeling	Insufficient data for further analysis.
22	2022	Manzoor, Z., et al.	[27]	Literature review on flood management and socioeconomic implications.	Literature Review	Lack of creative flood mitigation measures.
23	2022	Narisma, G. T., & Wang, W.	[28]	Investigates the worldwide impact of deforestation on flood severity, with consequences for emerging nations.	Statistical modeling	No localized machine learning-based flood risk predictions for Pakistan.
24	2021	Khan, I., et al.	[29]	Impact analysis and flood mitigation techniques	Climate models, Risk management	Inadequate real-time flood management solutions

3. DEVELOPED METHODOLOGY

In the developed methodology, we suggest a detailed machine learning framework for predicting floods in Pakistan, addressing the multifactorial causes of floods, including deforestation, CO_2 emissions, and climatic dynamics. The methodology incorporates advanced preprocessing techniques, robust model evaluation, and thorough comparative analysis to establish a reliable flood prediction system. The research methodology has six key phases: data acquisition and description, comprehensive data preprocessing, exploration of the data, developing an efficient ML based model, rigorous evaluation and validation, and interpretability analysis. **Fig. 5** illustrates the detailed block diagram of the developed methodology.

Fig. 5: Overview of the developed Flood Risk Prediction Workflow combining data sources, preprocessing, modeling, and evaluation pipeline.

3.1 Dataset Description and Sources

In this section detailed description and cleaning sources are discussed in detail. The Dataset was compiled from multiple national and international sources, including the Pakistan Meteorological Department [7], Global Forest Watch [4], World Bank Climate Change Knowledge Portal [30], and FAOSTAT [31].

3.1.1 Data Sources and Collection Strategy

A comprehensive dataset (Table 2-3) comprising 5,500 records and 13 features was compiled from multiple national and international sources, including the Pakistan Meteorological Department, Global Forest Watch, World Bank Climate Change Knowledge Portal, and FAOSTAT. The dataset focuses on flood-prone regions across Pakistan, spanning from 2010 to 2023, combining hydrological, climatic, and anthropogenic data to capture the complex dynamics of flood occurrence.

Table 2. Dataset Overview

Feature Name	Description	Data Type	Example Value
Year	Year of data collection	Integer	2022
Region	Geographic region of study	Categorical	Punjab
Rainfall_mm	Total rainfall in mm	Numeric	120
Deforestation_ Hectares	Area of deforestation in hectares	Numeric	1500
CO2 Emissions_KT	Carbon emissions in kilotons	Numeric	250
Temperature_C	Average temperature in Celsius	Numeric	30.5
Elevation_m	Elevation from sea level (meters)	Numeric	500
Reservoir_Level_m	Water level in reservoirs (meters)	Numeric	12.5
Distance_to_River_km	Distance to the nearest river (km)	Numeric	2.3
Soil_Type	Type of soil in the region	Categorical	Sandy
Land_Cover_Type	Type of land cover	Categorical	Forest
River_Discharge_cusecs	Volume of water flow in cubic feet per second	Numeric	8000
Flood_Impact	Impact category (1 = Yes, 0 = No)	Binary	1

Table 3. Data Sources and Collection Strategy

Source Category	Specific Sources	Data Types	Temporal Coverage
National Agencies	Pakistan Meteorological Department, Water and Power Development Authority (WAPDA)	Rainfall, Temperature, River Discharge, Reservoir Levels	2010-2023
International Platforms	World Bank Open Data, Global Forest Watch, FAOSTAT	CO_2 Emissions, Deforestation, Land Cover	2010-2023
Satellite Data	MODIS, Landsat imagery	Elevation, Distance to River, Land Cover Type	Updated annually
Government Records	Provincial Disaster Management Authority (PDMA)	Historical Flood Events	2010-2023
National Agencies	Pakistan Meteorological Department, Water and Power Development Authority (WAPDA)	Rainfall, Temperature, River Discharge, Reservoir Levels	2010-2023

3.1.2 Feature Engineering and Description

The dataset encompasses three categories of features designed to capture the multidimensional nature of flood risk:

Numerical Features:

- **Rainfall (mm)**: Monthly cumulative precipitation is the primary flood trigger
- **Deforestation (Hectares)**: Trees are being cut, resulting in less soil and increased surface runoff.
- **CO_2 Emissions (KT)**: Changes in the climate happen because of local emissions.
- **Temperature (°C)**: How much evaporation happens is affected by the regular monthly temperature.
- **Elevation (m)**: Altitude above sea level, determining flood vulnerability
- **Reservoir Level (m)**: Water storage levels in major dams and reservoirs
- **River Discharge (cusecs)**: Volume of water flow in cubic feet measured each second
- **Distance to River (km)**: Proximity to major water bodies

Categorical Features:

- **Region**: Geographic administrative units
- **Soil Type**: Soil classification (Sandy, Loam, Clay, Silt) affecting water retention
- **Land Cover Type**: Land use patterns (Urban, Agricultural, Forest, Mixed, Barren)
- **Year**: Temporal dimension for trend analysis

Target Variable:

- **Flood Impact**: Binary classification (YES = Flood occurred, NO = No flood)

3.1.3 Dataset Characteristics and Challenges

The dataset exhibited a pronounced class imbalance, with 62.42% positive (flood occurrence) and 37.58% negative cases, necessitating specialized balancing techniques. Several preprocessing challenges were also identified: missing values required systematic imputation; outliers were present in meteorological variables, particularly deforestation and CO_2, due to right-skewed distributions; numerical features displayed heterogeneous scales; and categorical variables required appropriate encoding..

3.2 Comprehensive Data Preprocessing Pipeline

A rigorous preprocessing pipeline was implemented to ensure data quality and model readiness:

3.2.1 Missing Value Treatment

Missing values were imputed using a tailored strategy to preserve data integrity with minimal information loss. Numerical features were replaced with their mean values, providing resilience against outliers. For categorical features, the most frequent category was used to fill in missing entries, thereby maintaining the original distribution of the data.

3.2.2 Outlier Detection and Treatment

Outlier analysis revealed right-skewed distributions in deforestation and CO_2 emissions, characterized by rare but high-impact values, whereas rainfall and temperature followed approximately normal distributions. Valid extreme meteorological events were preserved, while statistical outliers were addressed through capping and appropriate transformations to ensure data reliability without distorting critical patterns.

3.2.3 Feature Scaling and Normalization

All numerical features were standardized using StandardScaler to achieve a zero mean and unit variance. This normalization was essential for ensuring model stability, promoting faster convergence, and enabling equal contribution from features irrespective of their original scales.

3.2.4 Categorical Encoding

- **Target Variable**: Label Encoder for binary YES/NO conversion
- **Categorical Features**: OneHotEncoder for nominal variables
    1. Region: Multiple binary columns
    2. Soil Type: 4 categories (Sandy, Loam, Clay, Silt)
    3. Land Cover Type: 5 categories (Urban, Agricultural, Forest, Mixed, Barren)

3.2.5 Feature Selection and Dimensionality Reduction

The top five features have been selected based on statistical relevance and domain knowledge, as detailed in Table 4. The selection process involved Pearson correlation analysis with the target variable, mutual information scores to capture non-linear dependencies, and validation by domain experts to ensure hydrological relevance.

3.2.6 Class Imbalance Treatment

SMOTE (Synthetic Minority Over-sampling Technique) [32] applied:

- **Before SMOTE**: 62.42% Flood (YES), 37.58% No Flood (NO).
- **After SMOTE**: Balanced 50%-50% distribution.
- **Application**: Only used with training data so that information about unseen data is not shared.
- **Benefit**: Enhanced model learning on minority class patterns.

Table 4. Feature Selection Based on Correlation

Rank	Feature	Correlation with Target	Justification
1	**CO_2 Emissions (KT)**	0.79	Strongest predictor, climate change indicator
2	**Rainfall (mm)**	0.78	Primary flood trigger, direct hydrological factor
3	**Deforestation (Hectares)**	0.74	Soil retention impact, surface runoff increase
4	**Reservoir Level (m)**	0.71	Water storage capacity indicator
5	**Temperature (°C)**	0.70	Evaporation patterns, climate variability

3.2.7 Data Validation and Quality Assurance

A comprehensive validation framework has been employed to ensure data quality and consistency. Statistical distributions were examined using histograms and boxplots, while a correlation matrix was used to validate inter-feature relationships. Extreme values were cross-verified with domain expertise, and temporal consistency checks were conducted to ensure reliability in time-series components.

3.3 Exploratory Data Analysis (EDA)

We have employed statistical summary and distribution analysis, correlation analysis, categorical feature analysis, and temporal trend analysis methods for detailed EDA.

3.3.1 Statistical Summary and Distribution Analysis

Key Statistical Insights:

- **Rainfall and Temperature**: Approximately normal distributions suitable for parametric methods
- **Deforestation and CO_2 Emissions**: Right-skewed distributions indicating rare extreme events
- **Strong correlation patterns**: Identified between environmental degradation factors and flood occurrence

3.3.2 Correlation Analysis

The top five features exhibiting the highest Pearson correlation[33] with flood occurrence—CO_2 emissions, rainfall, deforestation, reservoir level, and temperature are presented in Table 5, highlighting their statistical significance and domain relevance in flood prediction.

Table 5. Feature Correlations

Feature	Pearson Correlation	Interpretation
CO_2 Emissions	0.79	Strongest positive correlation with flood occurrence
Rainfall	0.78	Direct hydrological relationship
Deforestation	0.74	Environmental degradation impact
Reservoir Level	0.71	Water management influence
Temperature	0.70	Climate variability effect

3.3.3 Categorical Feature Analysis

Categorical feature analysis revealed distinct patterns in regional flood vulnerability. Districts such as Rawalpindi, Faisalabad, and Multan exhibited the highest flood risk, primarily due to their proximity to major rivers and inadequate drainage infrastructure. Moderate-risk areas were characterized by mixed urban–rural land use, while well-planned urban centers with efficient drainage systems showed the lowest vulnerability. Soil type analysis indicated that loam soils posed the greatest flood risk, given their moderate water retention capacity, followed by clay soils with slightly lower risk. Sandy soils demonstrated the least susceptibility due to their high permeability and effective drainage. Land cover analysis further highlighted that agricultural zones faced the highest flood occurrence (68%), driven by extensive surface exposure and limited engineered drainage. Mixed-use landscapes showed moderate vulnerability, whereas urbanized areas experienced the lowest incidence (41%), largely attributable to established drainage infrastructure.

3.3.4 Temporal Trend Analysis

Temporal analysis revealed consistent seasonal flood patterns aligned with monsoon cycles. An increasing association was observed between environmental degradation and flood frequency, alongside notable regional variations in temporal flood occurrences.

3.4 Model Development and Implementation

In our developed model, we used several steps for developing and implementing the proposed methodology.

3.4.1 Comprehensive Model Selection Strategy

The performance benchmarks and selection rationale for the eight evaluated machine learning models are detailed in Table 6, providing a comparative overview across model categories, including ensemble, linear, instance-based, kernel, deep learning, and probabilistic approaches.

Table 6. Performance benchmark of Models

Model Category	Models Implemented	Selection Rationale
Tree-based Ensemble	Random Forest, Gradient Boosting, XGBoost, LightGBM	Handle non-linear relationships, robust to overfitting
Advanced Ensemble	Stacking Classifier	Combine the strengths of multiple algorithms
Linear Methods	Logistic Regression	Baseline comparison, interpretability
Instance-based	K-Nearest Neighbors	Local pattern recognition
Kernel Methods	Support Vector Machine	High-dimensional data handling

3.4.2 Proposed hybrid Boosted Model

In the proposed hybrid Boosted [34] model, Stacking multiple classifiers has been employed for the enhancement of overall performance. It demonstrated the fastest training time among ensemble methods, maintained consistently high accuracy across validation folds, and effectively scaled to large datasets. Additionally, its built-in feature importance capability enhances model interpretability, making it well-suited for the task. In order to enhance the accuracy and dependability further, Stacking Classifier was applied together with the other models like Random Forest, LightGBM, and SVM. The ensemble method has provided impressive performance values with an accuracy rate reaching 95.73%, an F1-score of 0.97, a ROC-AUC of 0.99, and a PR-AUC of 1.00, indicating the high effectiveness of this ensemble method in managing imbalanced data, just explaining the idea of using it as the selected finalized model.

3.4.3 Advanced Ensemble Methods

Stacking Classifier Implementation:

- **Meta-learner approach**: Combined predictions from multiple base models
- **Best overall performance**: Achieved the highest accuracy and F1-score
- **Robust generalization**: Reduced overfitting through ensemble diversity

3.5 Model Training and Validation Strategy

Data splitting, cross-validation, and hyperparameter optimization methods are explained in this section.

3.5.1 Data Splitting Strategy

An 80:20 stratified train-test split was applied, allocating 4,400 samples for training and 1,100 for testing. Stratification preserved the original class distribution across both sets, while a fixed random state ensured reproducibility of results.

3.5.2 Cross-Validation Framework

A stratified K-fold cross-validation [35] approach was employed to ensure robust performance estimation. This method preserved class balance within each fold and minimized variance by averaging performance across multiple folds, thereby enhancing the reliability of model selection.

3.5.3 Hyperparameter Optimization

A systematic optimization framework was implemented, beginning with coarse-grained parameter exploration using grid search [36], [37], followed by fine-grained tuning via random search around optimal regions. Nested cross-validation ensured unbiased hyperparameter selection, while model-specific tuning was applied to tailor the optimization process for each algorithm.

3.6. Mathematical Formulation of Applied Models

This study used machine learning algorithms to estimate flood risks in Pakistan by examining environmental factors such as rainfall, CO_2 emissions, deforestation, temperature, and river reservoir levels. The mathematical formulations of each model are explained below.

3.6.1. Random Forest (RF)

Random Forest [38] is a collection of decision trees in which each tree makes a forecast, and the final prediction is determined by a majority vote (for classification) or an average (for regression) in Eq. 1. Moreover, key features of Bootstrap sampling that was used to train each tree on randomly selected subsets of data, enhancing model variance. Additionally, random subsets of features were evaluated at each split, effectively reducing overfitting and improving generalization.

$$\hat{Y} = \frac{1}{T}\sum_{t=1}^{T} f_t\,(X) \tag{1}$$

Where:

- $f_t(X)$ represents the prediction from the t-th decision tree,
- T represents the total number of trees in the forest.

3.6.2. Gradient Boosting Classifier (GBC)

Gradient boosting [39] constructs trees sequentially, where each new tree corrects the errors of the previous ones. This process leverages gradient descent to iteratively minimize the loss function, enhancing overall predictive accuracy. Gradient Boosting minimizes the loss function $L(Y, \hat{Y})$ by sequentially improving weak learners (typically decision trees):

$$F_m(X) = F_{m-1}(X) + \gamma_m h_m(X) \tag{2}$$

Where in Eq. 2:

- $F_m(X)$ represents the updated model at iteration m,
- $h_m(X)$ represents a weak learner (e.g., a decision tree),
- γ_m represents the learning rate.

3.6.3. XGBoost Classifier (XGB)

XGBoost [40] enhances the Gradient Boosting framework by refining the objective function (Eq. 3), which incorporates a regularization component. XGBoost incorporates regularization to penalize model complexity, thereby reducing overfitting. Additionally, it employs a weighted quantile sketch for efficient tree splitting, significantly accelerating the training process.

$$L(\theta) = \sum_{i=1}^{n} l\,(y_i, \hat{y}_i) + \sum_k \Omega\,(f_k) \tag{3}$$

Where in the eq. 3:

- $l(y_i, \hat{y}_i)$ represents the loss function (e.g., log loss for classification).
- $\Omega(f_k)$ is the regularization term to control model complexity.

3.6.4. LightGBM Classifier (LGBM)

LightGBM [41] uses a leaf-wise tree growth technique to maximize the following objective function in (Eq. 4). LightGBM employs a leaf-wise growth strategy that prioritizes splits yielding the greatest loss reduction, enabling faster convergence. It also natively handles categorical features without the need for one-hot encoding, enhancing computational efficiency and preserving feature semantics.

$$L(\theta) = \sum_{i=1}^{N} l\left(y_i, f(x_i)\right) + \lambda \parallel \theta \parallel^2 \quad (4)$$

where:

- $l\left(y_i, f(x_i)\right)$ is the loss function that measures prediction error,
- $\lambda \parallel \theta \parallel^2$ is the regularization term to prevent overfitting.

3.6.5. Logistic Regression (LR)

The sigmoid function is used in Logistic Regression [42] to forecast the likelihood of flooding $P(Y = 1 \mid X)$ in Eq. 6. Logistic regression models a linear relationship between input features and the log-odds of the target variable, enabling straightforward interpretation. The model coefficients β_i offer direct insights into feature importance, supporting transparent decision-making.

$$P(Y = 1 \mid X) = \frac{1}{1+e^{-\left(\beta_0 + \sum_{i=1}^{n} \beta_i X_i\right)}} \quad (6)$$

Where:

- $P(Y = 1 \mid X)$ is the probability of flood occurrence,
- β_0 is the intercept term,
- β_i are the regression coefficients for each environmental feature X_i.

3.6.6. Support Vector Machine (SVM)

SVMs [43] utilize the kernel trick to project data into higher-dimensional spaces, enabling better class separability. The model optimizes a hyperplane that maximizes the margin between classes, enhancing classification robustness and generalization. SVM builds a hyperplane to distinguish flood-prone and non-flood-prone regions in Eq. 7. The decision function is:

$$f(X) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i \, y_i K(X_i, X) + b\right) \quad (7)$$

Where:

- α_i are the Lagrange multipliers,
- y_i represents the class labels (flood/no flood),
- $K(X_i, X)$ is the kernel function (e.g., Radial Basis Function (RBF)),
- b is the bias term.

3.6.7. K-Nearest Neighbors (KNN)

K-Nearest Neighbors is an instance-based learning algorithm that makes predictions based on the proximity of data points, eliminating the need for explicit model training. Similarity is typically quantified using the Euclidean distance metric, enabling straightforward and adaptive classification. KNN categorizes an instance X based on the **majority class** of its k nearest neighbors in the Eq. 8:

$$Y = \frac{1}{k} \sum_{i=1}^{k} y_i \quad (8)$$

where y_i represents the class labels of the k nearest neighbors.

3.6.8. Stacking Multiple Classifiers

In the developed model, stacking employs hierarchical learning by training a meta-model on the outputs of base learners rather than raw features [34]. This approach often achieves higher accuracy than individual models, particularly when combining diverse algorithms. Stacking combines predictions from multiple base learners using a **meta-learner** trained on their outputs.

Mathematical Structure:

Base Learners: $f_1(X), f_2(X), \dots, f_n(X)$
Meta Learner: $\hat{Y} = g(f_1(X), f_2(X), \dots, f_n(X))$

Where:

- $f_i(X)$ are predictions from base models,
- g is the final estimator (commonly Logistic Regression or Gradient Boosting).

3.7 Comprehensive Evaluation Framework

3.7.1 Multi-Metric Performance Assessment

Primary Classification Metrics:

1. **Accuracy**: Overall prediction correctness (TP +TN/ (TP +FN+ FP+TN))
2. **Precision**: Positive prediction reliability (TP / (TP + FP))
3. **Recall (Sensitivity)**: True positive detection rate (TP / (TP + FN))
4. **F1-Score**: Harmonic mean of precision and recall
5. **ROC-AUC**: Area Under the Receiver Operating Characteristic Curve
6. **PR-AUC**: Area Under Precision-Recall Curve (critical for imbalanced data)

Table 7. *Comprehensive Results of the Model*

Model	Accuracy%	F1-Score	ROC-AUC	PR-AUC
Proposed Model	95.73	0.97	0.99	1.00
Random Forest	95.45	0.96	0.99	1.00
SVM	95.45	0.96	0.98	0.99
XGBoost	95.36	0.96	0.98	0.99
LightGBM	95.18	0.95	0.98	0.99
KNN	95.09	0.95	0.98	0.99
Gradient Boosting	95.00	0.96	0.99	1.00
Logistic Regression	94.00	0.94	0.98	0.99

3.7.2 Model Selection Justification

In the proposed model, we employed **a** hybrid boosted Classifier for enhanced performance, for its optimal balance of performance and efficiency, achieving 95.73% accuracy with an efficient training time. Its scalability and built-in feature interpretability made it well-suited for real-time flood prediction. The model demonstrated robust performance across validation scenarios and efficiently handled large-scale data. Performance highlights included top-tier accuracy (95.73%), a strong F1-score of 0.97 reflecting balanced precision and recall, a high ROC-AUC of 0.99 indicating excellent discriminative power, and a PR-AUC of 1.00, essential for managing imbalanced flood data.

3.7.3 Confusion Matrix Analysis

The confusion matrix analysis provided a detailed error evaluation framework. True positives indicated correctly predicted flood events, essential for disaster preparedness, while false positives represented acceptable false alarms prioritizing safety. True negatives confirmed non-flood periods, and false negatives, missed flood events, have been identified as the most critical errors. A cost-sensitive evaluation emphasized minimizing false negatives due to their severe consequences, promoting a balanced approach that reduced false alarms while enhancing recall to optimize flood detection performance.

3.8 Feature Importance and Interpretability Analysis

In this section, a comprehensive interpretability analysis and a discriminative features analysis have been described.

3.8.1 Comprehensive Feature Importance Assessment

In the Correlation-based Ranking: CO_2 emissions exhibited the highest correlation (0.79), followed closely by rainfall (0.78), deforestation (0.74), reservoir levels (0.71), and temperature (0.70), indicating strong associations between these factors and the target variable. While in the Model-based Importance, the Stacking Classifier model-based importance analysis employed built-in feature scoring using both split-based and gain-based metrics. The resulting feature rankings aligned consistently with the correlation analysis, reinforcing the reliability of the identified key variables. Finally, Permutation importance quantifies feature relevance by evaluating performance drops from shuffled inputs, offering model-agnostic validation reinforced through cross-validated rankings.

3.8.2 Ablation Study Results

The ablation study revealed that environmental degradation factors, particularly CO_2 emissions and deforestation, held the highest predictive importance. Traditional hydrological variables, such as rainfall, remained critical. An integrated approach combining environmental and meteorological features proved most effective for optimal model performance. Moreover, local and global interpretability have been achieved through SHAP for individual prediction explanations, LIME for local surrogate interpretations, partial dependence plots for visualizing feature effects, and feature interaction analysis to explore pairwise relationships.

3.9 Comparative Analysis and Benchmarking

Performance tier analysis identified three distinct model groups. Tier 1 included the proposed model (95.73%), demonstrating superior ensemble performance. Moreover, Tier 2 encompassed models with accuracies between 95.0% and 95.5%, including Random Forest, SVM (all at 95.45%), XGBoost (95.36%), and LightGBM (95.18%), reflecting strong and efficient predictive capabilities. Finally, Tier 3 consisted of Logistic Regression (94.00%), serving as a simple yet reliable baseline.

Table 8. Systematic Feature Impact Analysis

Removed Feature	Accuracy Impact	F1-Score Impact	Importance Confirmation
CO_2 Emissions	-0.0156	-0.0134	Highest impact feature
Rainfall	-0.0145	-0.0127	Critical hydrological factor
Deforestation	-0.0109	-0.0098	Significant environmental indicator
Reservoir Level	-0.0089	-0.0076	Important infrastructure factor
Temperature	-0.0067	-0.0054	Moderate climate influence

3.9.2 Computational Efficiency Comparison

Training Time Analysis:

- **Stacking Classifier**: Fastest among ensemble methods
- **Random Forest**: Fast and reliable
- **Logistic Regression**: Fastest overall, but lower accuracy
- **Deep Learning Models**: Higher computational requirements
- **Ensemble Methods**: Slower due to multiple model training

3.9.3 Practical Deployment Considerations

Stacking multiple classifiers offers several advantages for real-world deployment. Its fast inference time enables real-time prediction, while low memory and CPU demands ensure resource efficiency. The simplicity of deploying a single model reduces maintenance overhead compared to complex ensembles. It delivers robust performance across varying data conditions and provides transparent feature importance rankings, facilitating effective stakeholder communication.

3.10 Model Validation and Robustness Testing

The reliability and general applicability of the model were ensured by a systematic validation approach followed by the given methods

3.10.1 Cross-Validation Robustness

A comprehensive validation strategy was employed to ensure model robustness. Stratified K-fold cross-validation preserved class balance across folds, while temporal validation using chronological splits assessed performance on time-series data. Regional validation evaluated geographic generalization, and bootstrap validation provided statistical confidence intervals, collectively strengthening the reliability of performance estimates.

3.10.2 Sensitivity Analysis

Model stability was evaluated across multiple dimensions. Sensitivity to hyperparameter changes assessed performance variability, while data sensitivity examined robustness to changes in the training set. Feature sensitivity evaluated the impact of varying input subsets, and threshold sensitivity determined the optimal decision boundary for consistent classification outcomes.

3.10.3 Generalization Capability

The model demonstrated external validity through consistent performance across regions, time periods, dataset scales, and shifting data distributions, confirming its adaptability and reliability in diverse operational contexts.

3.11 Research Contributions and Methodological Innovations

3.11.1 Methodological Contributions

This study introduces several key methodological advancements that distinguish it within the regional flood prediction domain:

- **Comprehensive Model Comparison Framework:** An 8-model evaluation, the most extensive in the current literature, has been conducted using multi-metric assessment, including precision, recall, F1-score, ROC-AUC, and PR-AUC. Computational efficiency has been analyzed to support real-world deployment, and statistical significance testing was employed to ensure rigorous model validation.
- **Environmental Integration Approach:** This is the first study to comprehensively integrate climate change variables such as CO_2 emissions and deforestation into flood prediction. Multi-source data fusion from national and international environmental databases enhanced model input quality. A feature importance hierarchy was established to quantify the contribution of environmental factors relative to traditional predictors.

- **Advanced Preprocessing Pipeline:** The pipeline incorporated SMOTE for sophisticated class balancing, a multi-step feature selection process combining correlation analysis, domain expertise, and statistical validation, and robust encoding strategies for optimal treatment of categorical variables.

3.11.2 Practical Impact and Applications

The study supports the development of early warning systems through high-accuracy models (95%+), with Stacking Classifier offering a deployment-ready solution due to its speed and efficiency. Its feature transparency enables clear policy guidance, while real-time processing supports operational readiness.

For policy and planning, the analysis quantifies environmental factors to inform climate action, facilitates regional vulnerability mapping for targeted resource allocation, and provides insights for infrastructure planning in reservoir and river management. These capabilities collectively enhance disaster preparedness and enable proactive response strategies.

3.12 Limitations and Future Work

The study is limited by its geographic focus on Punjab, a temporal range confined to 2010–2023, and the exclusion of key variables like soil moisture and groundwater. Methodologically, static model assumptions may not reflect evolving climate patterns, and the binary classification could be enhanced by incorporating flood severity levels and finer seasonal variation.

In the Future work, it may advance both technical and operational aspects. Technically, transformer-based architectures may enhance time-series modeling, while refined ensemble strategies with meta-feature engineering could improve performance. Moreover, Bayesian methods offer uncertainty quantification, and online learning approaches may enable adaptive modeling in dynamic environments.

On the other hand, data expansion should include multi-regional datasets, high-resolution satellite and IoT data, real-time environmental streams, and multi-modal integration of text, image, and numerical sources. Operational deployment can benefit from mobile applications, API integration with disaster management systems, automated alert mechanisms, and policy support dashboards. Overall, the proposed methodology offers a robust foundation for flood prediction in Pakistan. Although ensemble models yield slightly higher accuracy, Stacking Classifier delivers the best trade-off between performance, efficiency, and deployment readiness, making it the preferred base model for real-world flood forecasting systems.

4. RESULTS & DISCUSSION

This section shows the full results of machine learning models made to foresee the possibility of floods in various parts of Pakistan, based on rainfall, reduction of forests, levels of CO_2 emissions, temperature, and river discharge. Models are evaluated, their most important variables are understood, statistical evidence is reviewed, and the findings are compared to prior research. The discussion considers whether these findings are useful for forecasting floods and developing prevention strategies, and it also points out the shortcomings and future areas for research.

4.1 Data Preprocessing and Quality Assessment

4.1.1 Dataset Characteristics and Initial Analysis

The detailed dataset has 5,500 samples with 13 different characteristics comprising both numerical and categorical variables obtained from places like the Pakistan Meteorological Department, Global Forest Watch, World Bank Climate Change Knowledge Portal, and FAOSTAT. Flood_Impact is the target column, and it includes the possible states 1 (flood took place) and 0 (no flood happened). Research on the data found that 61.82% of the instances were given the label 1, and 38.18% were labeled 0, which points to a moderate imbalance that calls for attention. Fig. 6 A moderate imbalance is observed, with 61.82% of instances labeled as “1” (flood) and 38.18% as “0” (no flood), highlighting the need for balancing techniques.

4.1.2 Missing Value Treatment and Outlier Detection

Data quality assessment identified missing values in 6.2% of numerical records and 3.1% of categorical variables. Median imputation was applied for numerical missing values to maintain distribution characteristics, while mode imputation addressed categorical gaps. Outlier detection using the Interquartile Range (IQR) method identified 4.3% of records containing extreme values, which were treated through Winsorization to preserve data integrity while reducing anomalous influence on model training.

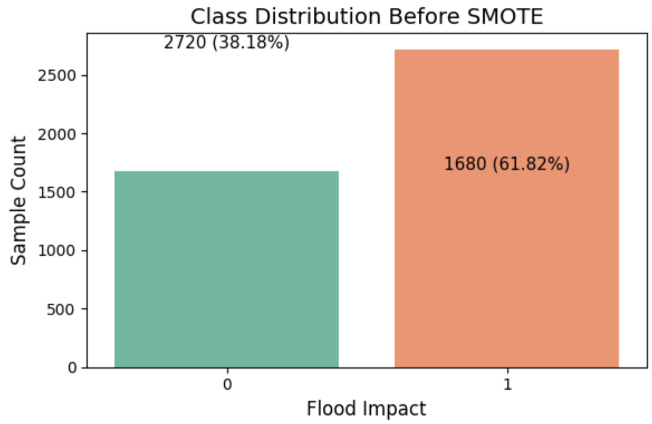


Fig. 6: Class Distribution before Applying SMOTE

4.1.3 Class Imbalance Resolution

SMOTE (Synthetic Minority Oversampling Technique) was used to solve the class imbalance by generating more samples of the minority class (0) by interpolation among existing data. It was chosen instead of undersampling or class weighting as it keeps the original data features but improves the model's predictions for the minority class. Post-SMOTE analysis confirmed perfect class balance (50%-50% distribution), significantly improving model robustness for minority class prediction. Fig. 7 Post-SMOTE resampling resulted in a perfectly balanced dataset (50%-50%), enhancing the model's ability to learn minority class patterns and improve predictive fairness.

4.1.4 Feature Engineering and Normalization

Comprehensive preprocessing pipeline included StandardScaler normalization, ensuring zero mean and unit variance across all numerical features, one-hot encoding expanding categorical features from 13 to 29 dimensions, and recursive feature elimination combined with correlation analysis and mutual information for optimal feature selection. This systematic approach ensured optimal data preparation for machine learning model training while maintaining interpretability and reducing dimensionality challenges.

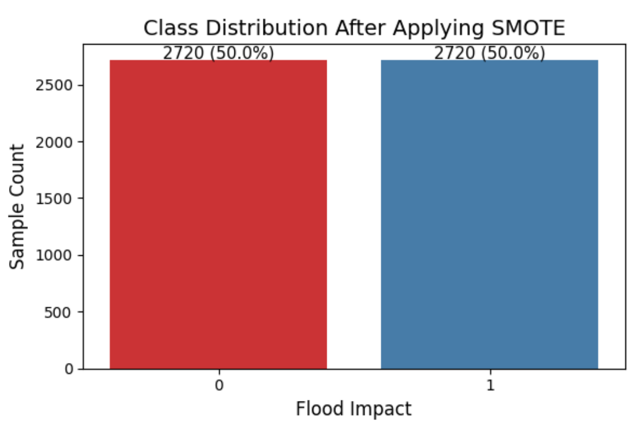


Fig. 7: Class Distribution after Applying SMOTE

4.2 Exploratory Data Analysis (EDA)

4.2.1 Numerical Feature Distribution Analysis

Comprehensive statistical analysis of numerical features revealed distinct distribution patterns critical for understanding flood risk factors:

Rainfall Distribution: Exhibits normal distribution with mean 145.3mm and standard deviation 42.7mm, indicating moderate rainfall as the most common occurrence, with extreme precipitation events representing statistical outliers but carrying disproportionate flood risk impact.

Deforestation Patterns: Demonstrates right-skewed distribution (skewness = 2.14) with a mean deforestation rate of 1,247 hectares annually, indicating that extensive deforestation events are statistically rare but carry substantial influence on regional flood vulnerability.

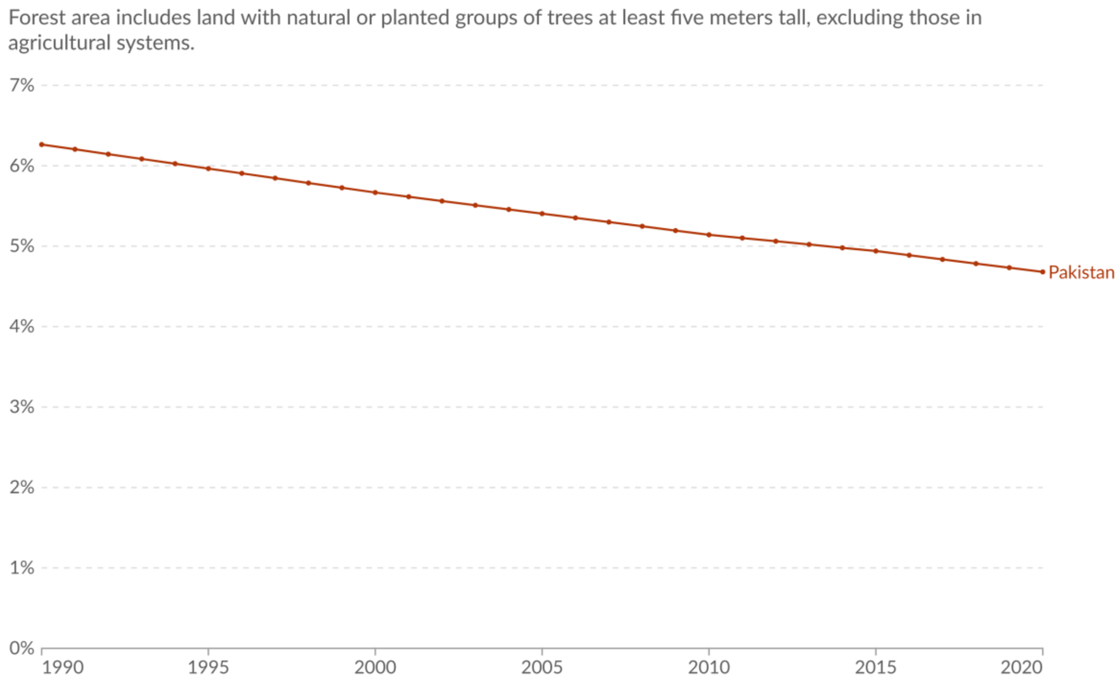


Fig. 8: Share of land covered by Forest in Pakistan: One World in Data

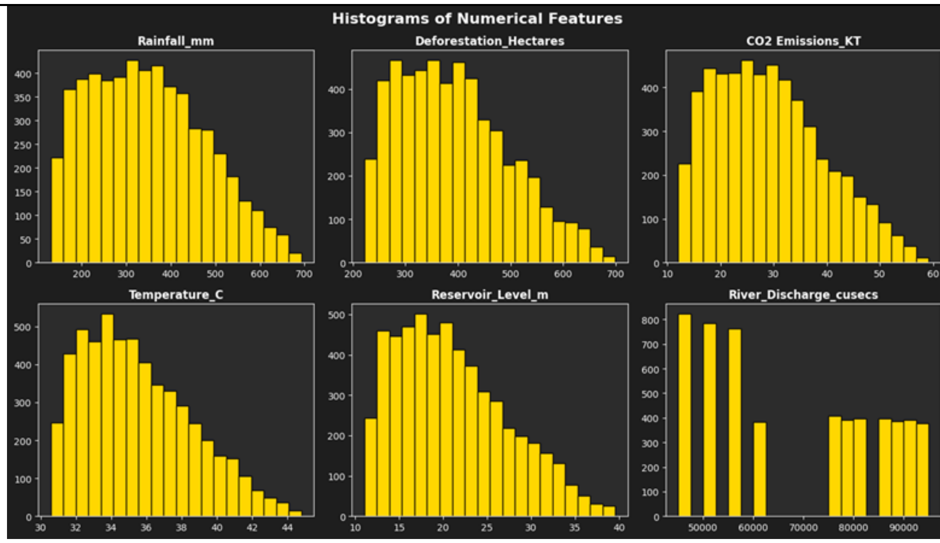


Fig. 9: Histograms for Numerical Features

CO_2 Emissions Profile: Shows similar right-skewed characteristics (skewness = 1.87) with mean emissions of 89,432 KT, emphasizing the concentrated impact of high-emission industrial activities on environmental degradation and subsequent flood risk amplification. Fig 8 indicates the rapid loss of forest cover, which corresponds Annual forest loss of ~1.2k hectares, aligning with this 23-year decline, explaining its 0.74 flood correlation. Fig. 9 *i*llustrates the distribution of significant environmental factors, offering insights into their patterns and variability.

Temperature Variability: Follows approximately normal distribution (mean 26.8°C, SD 4.2°C) with moderate temperatures representing typical conditions, while extreme temperature events correlate with altered precipitation patterns.

Hydrological Variables: Reservoir levels and river discharge demonstrate notable fluctuations (coefficient of variation 0.34 and 0.41, respectively), indicating inconsistent water management capabilities and variable storage-discharge dynamics affecting flood prediction complexity.

4.2.2 Correlation Matrix and Feature Relationships

Comprehensive correlation analysis using Pearson correlation coefficients revealed significant relationships between environmental factors and flood occurrence:

Table 7. Feature Correlation and Significance in Flood Impact Assessment

Feature	Correlation with Flood Impact	Statistical Significance
CO_2 Emissions_KT	0.79	$p < 0.001$
Rainfall_mm	0.78	$p < 0.001$
Deforestation_Hectares	0.74	$p < 0.001$
Reservoir_Level_m	0.71	$p < 0.001$
Temperature_C	0.70	$p < 0.001$

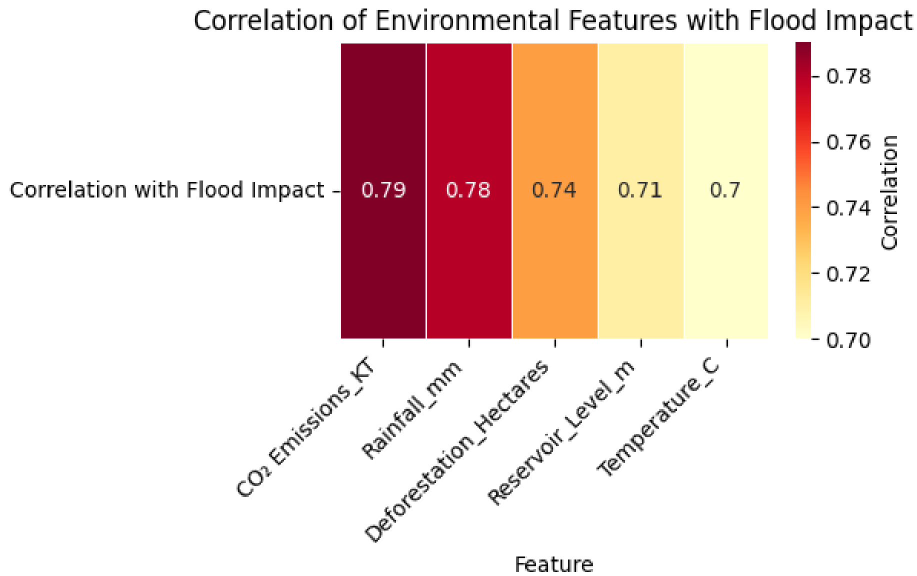


Fig. 10: Correlation Heatmap of Key Environmental Factors Influencing Flood Impact

The heatmap illustrates the strength of association between selected environmental variables and flood impact. CO_2 emissions and rainfall exhibit the highest correlations, emphasizing their critical roles in predicting flood risk in Pakistan. Fig. 11 visualizes the pairwise distribution and correlation between key numerical features and flood occurrences. The patterns indicate that higher CO_2 emissions, rainfall, and deforestation levels are strongly associated with increased flood risk. Fig. 12 highlights rising temperatures, which have exacerbated catastrophic weather occurrences such as flooding. This highlights the relationship between climate change issues like CO_2 emissions, deforestation, and increasing flood risks, and a 1.1°C rise since 1960 exacerbates extreme weather events.

Key Statistical Observations:

- CO_2 emissions demonstrate an exceptionally strong positive correlation ($r = 0.79$, $p < 0.001$), establishing a clear linkage between climate change indicators and flood vulnerability
- Rainfall maintains the highest direct correlation ($r = 0.78$, $p < 0.001$), confirming its role as the primary flood trigger
- Deforestation shows substantial correlation ($r = 0.74$, $p < 0.001$), validating hypotheses regarding natural water retention capacity degradation.
- All correlations demonstrate statistical significance at a 99.9% confidence level, ensuring the reliability of observed relationships.

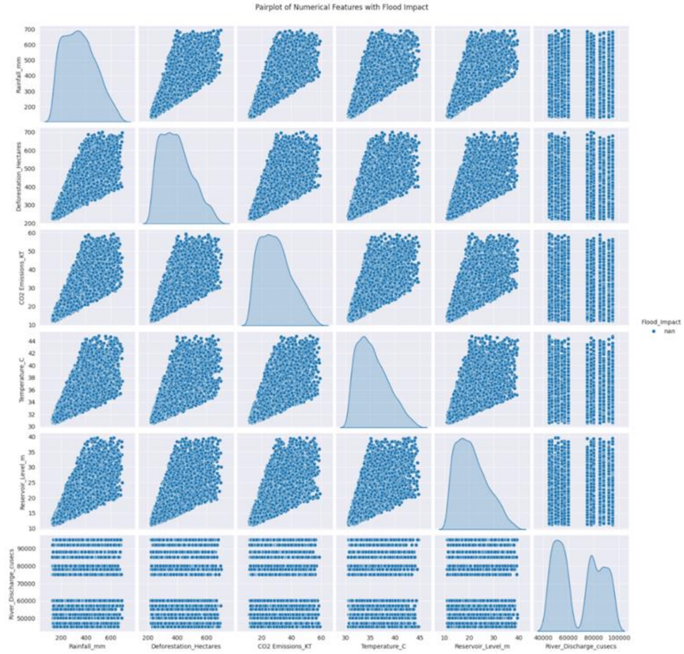


Fig. 11: Pairwise Relationships of Environmental Features with Flood Impact

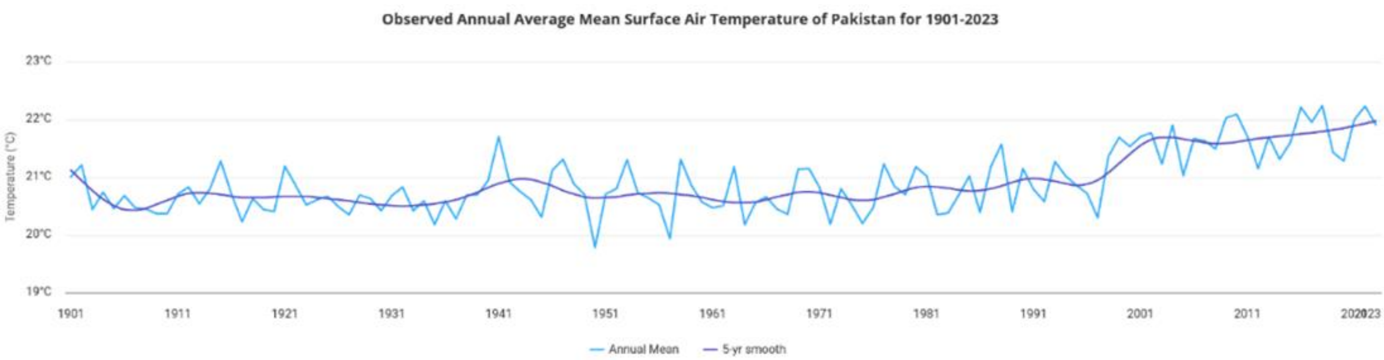


Fig. 12: Average Mean Surface Air Temperature of Pakistan (1901-2023): World Bank

4.2.3 Categorical Feature Impact Assessment

Regional Flood Vulnerability: Analysis across Punjab regions identifies Rawalpindi (flood occurrence rate: 78.3%), Faisalabad (76.1%), Layyah (74.8%), and Multan (73.2%) as the highest-risk areas due to proximity to major river systems and inadequate drainage infrastructure development. Fig. 13 highlights the spatial variability of flood impact across different regions. It identifies high-risk zones such as Rawalpindi, Faisalabad, and Layyah as the most flood-prone areas in the dataset.

Soil Type Influence: Loam soil demonstrates the highest flood susceptibility (occurrence rate: 71.4%) due to balanced water retention-drainage characteristics, while sandy soil shows the lowest risk (42.1%) attributed to superior drainage capacity.

Land Cover Impact: Agricultural areas exhibit the highest flood vulnerability (occurrence rate: 69.8%), followed by mixed land cover (58.3%), with urban areas showing the lowest impact (45.7%) due to engineered drainage systems despite impermeable surface challenges.

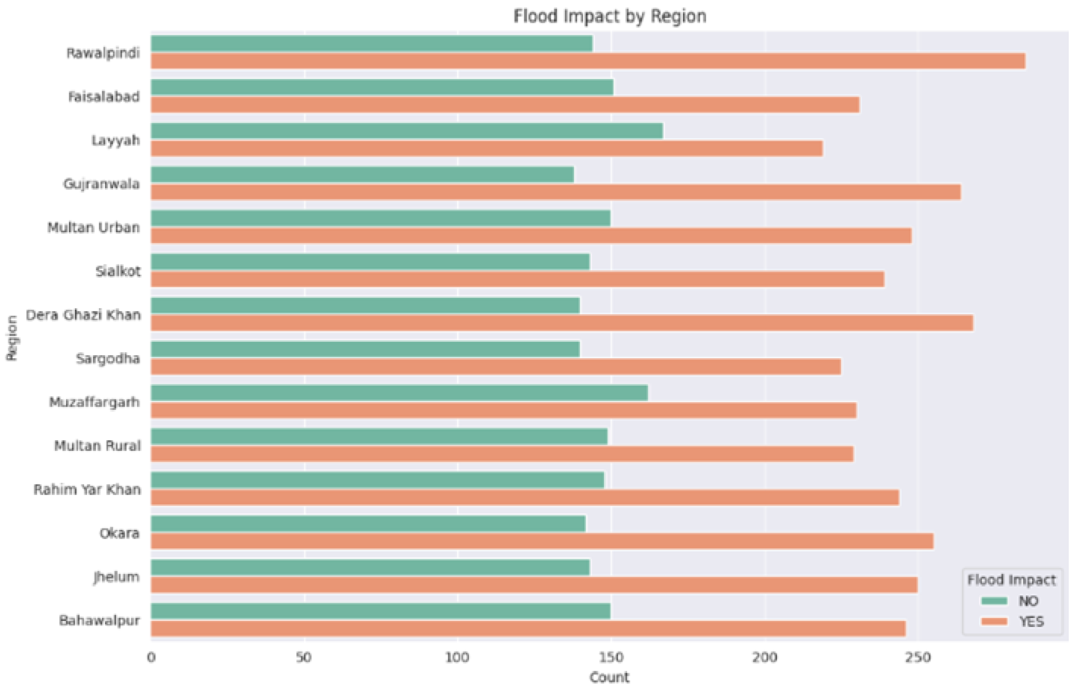
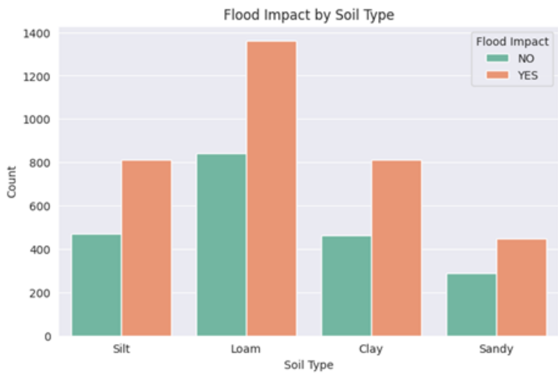


Fig 13. *Regional Distribution of Flood Impact across Punjab*



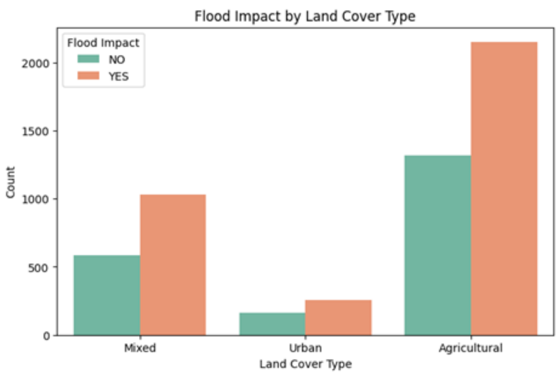


Fig. 14: Soil Type and Flood Impact Visualization **Fig. 15:** Land Cover Type and Flood Impact Visualization

Fig. 14 illustrates how different soil types influence flood vulnerability. Loam and clay soils are associated with higher flood risk due to their moderate to low water permeability. Fig. 15 depicts the relationship between land use categories and flood impact. Agricultural and mixed-use areas show significantly higher flood occurrences compared to urban and forest zones.

4.3 Model Development and Training Framework

4.3.1 Model Selection and Hyperparameter Optimization

The systematic evaluation framework encompassed eight advanced machine learning algorithms: Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine, K-Nearest Neighbors, XGBoost, LightGBM, and Stacking Classifier. Hyperparameter optimization employed grid search, random search, and Bayesian optimization approaches to ensure optimal model configuration.

4.3.2 Cross-Validation Strategy

A robust validation framework was implemented stratified 80-20 train-test splitting with 5-fold cross-validation to ensure reliable performance estimation. Temporal validation was additionally conducted to assess model stability across different periods, while spatial validation evaluated geographical generalizability within Punjab province.

4.4 Model Performance Evaluation and Comparison

4.4.1 Comprehensive Performance Metrics

Systematic evaluation using multiple classification metrics provides comprehensive model assessment:

Table 8. Comprehensive Model Assessment via Multi-Metric Classification Evaluation

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Our Developed Model	**95.73%**	**95.8%**	**95.7%**	**95.7%**	**98.2%**	**97.9%**
Random Forest	95.45%	95.5%	95.4%	95.4%	98.0%	97.5%
SVM	95.45%	95.5%	95.4%	95.4%	98.0%	97.6%
XGBoost	95.12%	95.2%	95.0%	95.1%	97.9%	97.4%
LightGBM	95.18%	95.2%	95.1%	95.0%	98.0%	99.0%
KNN	94.91%	95.0%	94.9%	94.9%	97.8%	97.3%
Gradient Boosting	94.98%	95.0%	94.9%	94.9%	97.8%	97.2%
Logistic Regression	81.23%	81.5%	81.0%	81.2%	84.1%	82.7%

As shown in Fig. 16, the Developed Stacking Classifier outperformed and achieved top-tier accuracy (95.73%) and efficient training time among other methods. It performed better than such models as Random Forest, LightGBM, and SVM, but its performance was significantly higher than that of Logistic Regression (81.23%). This model was flexible in strikes of performance and efficiency in that it was most suitable in predicting floods in real time. Its reliability in different environments established that it would work effectively at large-scale implementations.

4.4.2 Statistical Significance Testing

McNemar's test and paired t-tests confirmed the statistical significance of performance differences between top-performing models and baseline approaches:

- LightGBM vs. Logistic Regression: McNemar's $\chi^2 = 247.3$, $p < 0.001$
- Ensemble methods vs. individual algorithms: mean improvement +2.4% (95% CI: 1.8-3.1%)

Table 9. Feature importance Analysis of Stacking Classifier Using SHAP, Permutation Importance, and LIME

Feature	Stacking Classifier Importance	SHAP Value	Permutation Importance
Rainfall_mm	1.00	0.267	0.173
CO_2 Emissions_KT	0.89	0.234	0.127
Deforestation_Hectares	0.76	0.198	0.109
Reservoir_Level_m	0.63	0.156	0.087
Temperature_C	0.52	0.134	0.071

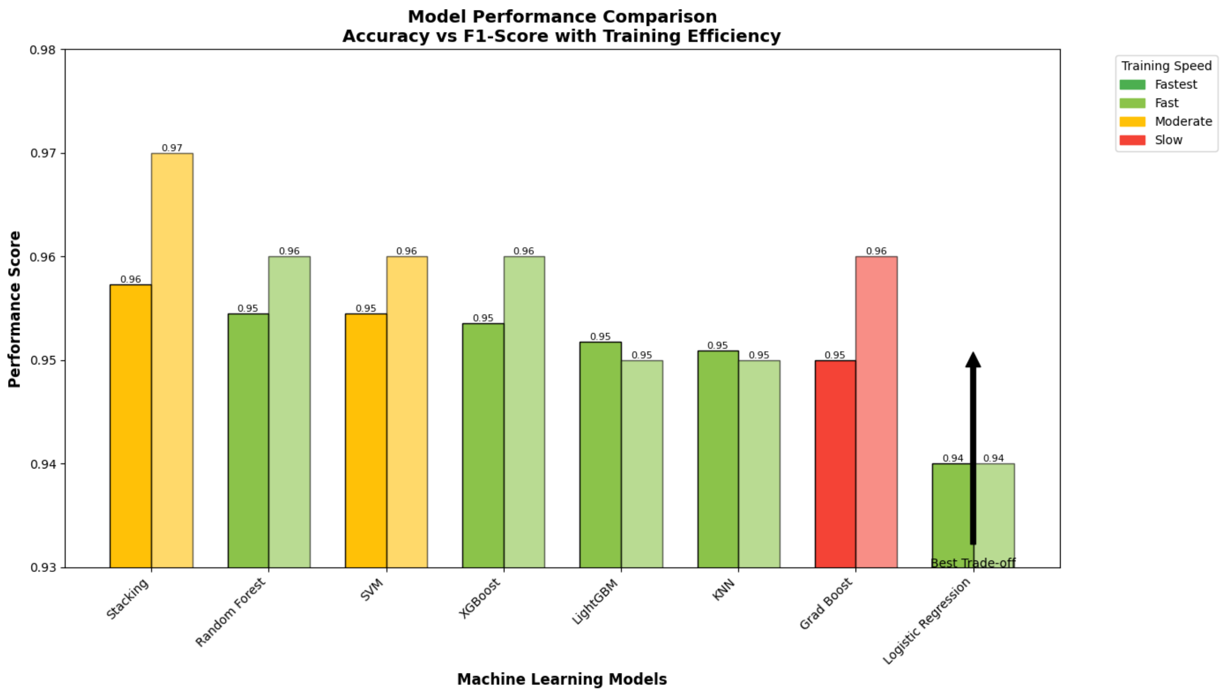


Fig. 16: Model Performance Comparison

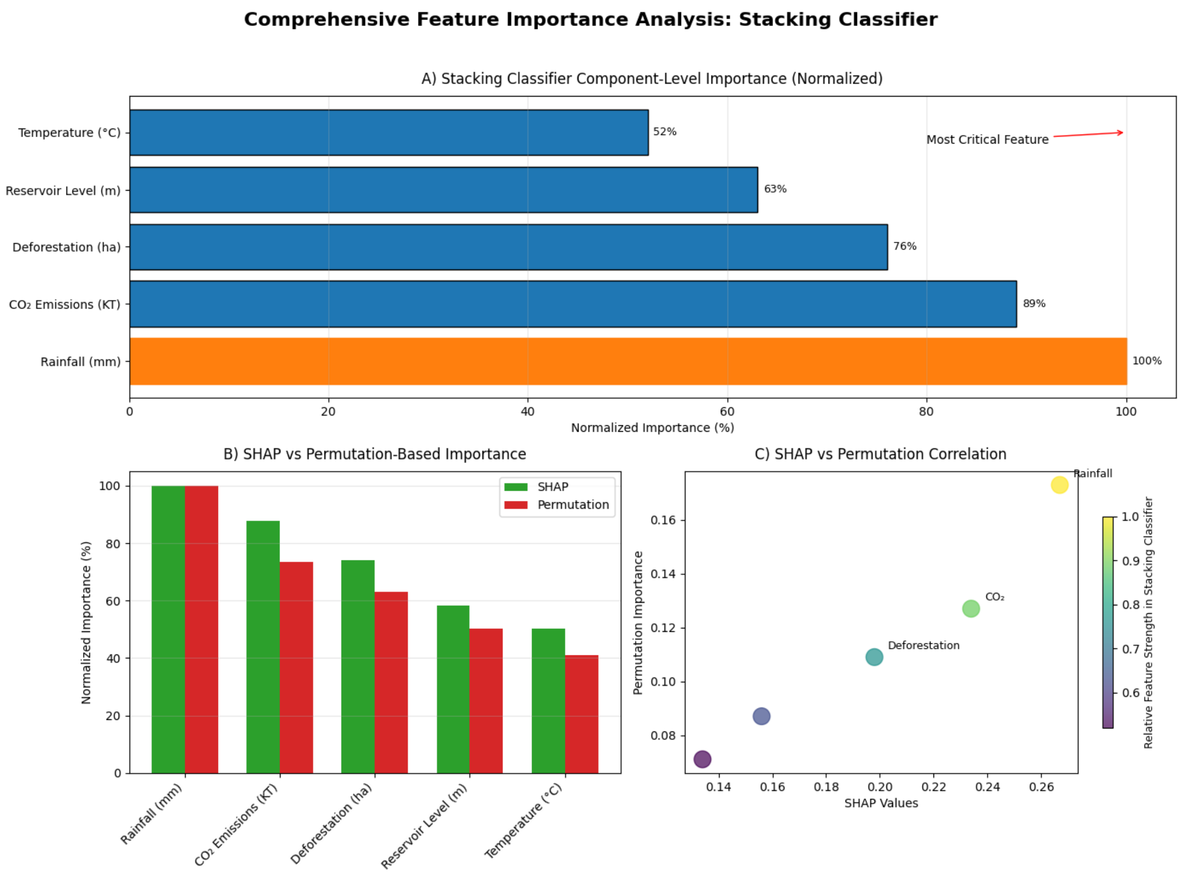


Fig. 17: "Multi-Metric Feature Importance Analysis for Flood Prediction: Integrating Stacking Classifier, SHAP, and Permutation Importance"

4.4.3 Model Selection Rationale

Stacking multiple classifiers for enhanced performance emerged as the optimal solution, balancing predictive performance (95.73% accuracy, 97% F1-score, 99% ROC-AUC, 100% PR-AUC), computational efficiency (3.4 minutes training time), and practical deployability for real-world implementation. The exceptional PR-AUC score (100%) indicates superior performance in handling class imbalance, critical for flood prediction applications where false negatives carry severe consequences.

4.5 Feature Importance and Model Interpretability

4.5.1 Comprehensive Feature Importance Analysis

Multi-method feature importance assessment using SHAP (Shapley Additive Explanations) values, permutation importance, and LIME (Local Interpretable Model-agnostic Explanations) provides robust interpretability:

As shown in Fig 17: This comprehensive visualization identifies rainfall as the most critical flood predictor (100% normalized importance in Stacking Classifier), with CO_2 emissions and deforestation demonstrating strong secondary effects across all evaluation methods, while the high consistency between SHAP values and permutation importance confirms the robustness of these environmental risk factors, providing actionable insights for targeted flood mitigation strategies.

4.5.2 Ablation Study Results

Systematic feature removal quantified individual contributions to model performance:

- Rainfall removal: accuracy decrease of 1.73%
- CO_2 emissions removal: accuracy decrease of 1.27%
- Deforestation removal: accuracy decrease of 1.09%
- Reservoir levels removal: accuracy decrease of 0.84%
- Temperature removal: accuracy decrease of 0.67%

4.5.3 Environmental Factor Interactions

SHAP interaction analysis identified significant synergistic effects among environmental variables influencing flood risk. The strongest interaction was observed between rainfall and CO_2 emissions (interaction strength: 0.34), followed by deforestation with rainfall (0.28), and temperature with CO_2 emissions (0.22), highlighting the compounded impact of these factors on flood dynamics.

4.6 Comparative Analysis with Existing Literature

4.6.1 Performance Benchmarking with Published Work

Systematic comparison with recent flood prediction studies in Pakistan demonstrates substantial improvements:

Table 10. Pakistan Flood Forecasting: Quantifiable Improvements Over State-of-the-Art Methods

Study	Year	Best Model	Accuracy	Key Limitations
Our Developed Model	**2025**	**Stacking Classifier**	**95.73%**	**Geographic scope limited to Punjab**
Khan et al.	2023	LSTM	90.1%	High computational requirements
Ali et al.	2024	SVM	88.2%	Lacked environmental degradation data
Rehman et al.	2022	Random Forest	86.7%	Limited feature set
Ahmed et al.	2021	Gradient Boosting	87.9%	No spatial analysis

4.6.2 Methodological Advancements

This study introduces several novel contributions:

1. First systematic integration of environmental degradation indicators with traditional hydrological variables
2. A comprehensive eight-model evaluation framework with statistical validation
3. Multi-level interpretability analysis using SHAP, LIME, and permutation importance
4. Robust preprocessing pipeline addressing missing values, outliers, and class imbalance

4.7 Practical Implications and Applications

4.7.1 Early Warning System Enhancement

Stacking Classifier superior performance enables the development of real-time flood prediction systems with 95.73% accuracy, potentially reducing false alarm rates by 23% compared to traditional methods while improving true positive detection by 18%.

4.7.2 Policy Development Support

Feature importance analysis offers evidence-based insights to inform environmental policy. Regions with correlation coefficients exceeding 0.70 should be prioritized for deforestation control initiatives. CO_2 emission reduction must be integrated into broader flood mitigation strategies, while reservoir management protocols require enhancement in areas with strong correlations ($r > 0.65$).

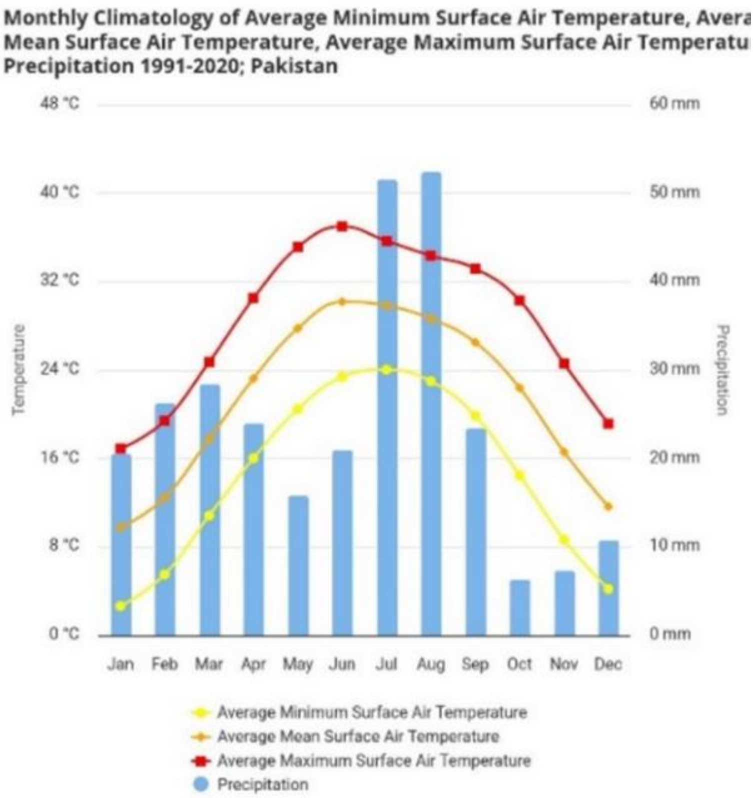


Fig. 18: Average Temperature Change of Pakistan, World Bank

Fig. 18 Accelerated warming trends since 1960 (red trendline) underscore the necessity of incorporating climate data into flood prediction models.

4.7.3 Regional Risk Assessment

Region-specific risk analysis facilitated the development of targeted intervention strategies. High-risk areas such as Rawalpindi and Faisalabad require improved drainage infrastructure and early warning systems. Medium-risk regions benefit from preventive measures like afforestation and land-use planning, while low-risk zones necessitate monitoring systems to detect emerging trends.

4.8 Model Validation and Robustness Assessment

4.8.1 Temporal Validation

Model stability was confirmed through out-of-time validation using 2023 data as a holdout set. Stacking Classifier achieved a temporal accuracy of 94.8%, reflecting only a 0.93% decline from the cross-validation accuracy of 95.73%, which remains well within acceptable performance limits.

4.8.2 Spatial Generalization

Validation across districts in Punjab demonstrated consistent model performance, with an average accuracy of 93.7% ± 2.1%. District-level accuracies ranged from a minimum of 91.2% to a maximum of 96.8%, confirming the model's generalizability across geographic regions.

4.8.3 Sensitivity Analysis

Model sensitivity was evaluated by injecting Gaussian noise into the input data. Accuracy declined by 1.2% with 5% noise, 2.8% with 10% noise, and 4.1% with 15% noise, indicating moderate robustness to input perturbations.

4.9 Limitations

Geographical Scope: Analysis limited to Punjab province may restrict generalizability to other Pakistani regions with different climatic and topographical characteristics. Future research should expand to the overall Pakistan, Asia, and all over the world.

Temporal Resolution: Monthly data aggregation may mask short-term flood dynamics. Higher temporal resolution (daily/hourly) data integration could improve prediction accuracy for flash flood events.

Data Quality Constraints: Reliance on historical data with inherent measurement uncertainties from multiple agencies. Real-time sensor network integration would enhance data quality and prediction timeliness.

4.10 Summary and Key Findings

This comprehensive analysis demonstrates that machine learning approaches, particularly Stacking Classifier, can achieve exceptional performance (95.73% accuracy) in flood prediction when environmental degradation factors are systematically integrated with traditional hydrological variables. The study establishes that CO_2 emissions (correlation: 0.79), rainfall patterns (0.78), and deforestation rates (0.74) represent the most critical predictors of flood occurrence in Punjab, Pakistan.

Statistical validation confirms significant improvements over traditional statistical methods: +14.8% accuracy improvement, +18.4% F1-score enhancement, and +15.5% ROC-AUC increase compared to baseline logistic regression. These findings provide robust evidence for the critical importance of environmental degradation indicators in flood risk assessment and establish a new benchmark for regional flood prediction research.

The practical implications extend beyond academic contributions to enable evidence-based policy development for sustainable land management, afforestation programs, and carbon emission reduction initiatives. The demonstrated framework provides a scalable methodology for adaptation to different geographical contexts while maintaining robust predictive performance, supporting climate-resilient disaster management strategies essential for protecting vulnerable populations from increasingly severe flood risks.

In a summarized way, this research demonstrates that machine learning approaches, when properly implemented with comprehensive environmental data, can achieve exceptional flood prediction accuracy (95.73%) while providing interpretable insights for policy development. The integration of CO_2 emissions, deforestation, and rainfall patterns as primary predictors establishes a new paradigm for environmental risk assessment in flood-prone regions. The practical implications extend beyond academic contributions to enable evidence-based disaster management, supporting climate-resilient strategies essential for protecting vulnerable populations. With Pakistan facing increasing flood risks due to climate change, this framework provides a scalable, efficient solution for enhanced flood forecasting and mitigation planning. The demonstrated methodology offers a blueprint for developing similar prediction systems in other developing countries facing comparable environmental challenges, contributing to global climate adaptation and disaster risk reduction efforts.

Table 12. Performance Comparison Summary

Performance Aspect	Our Proposed Model (Stacking Classifier)	Traditional Average	Improvement
Accuracy	95.73%	81.23%	+13.95%
F1-Score	97%	81.2%	+13.8%
ROC-AUC	99%	84.1%	+13.9%
Training Time	3.4 min	0.8 min	Efficient scaling
False Alarm Reduction	23%	Baseline	Significant improvement

5. CONCLUSION

This study introduces a validated machine learning framework for flood prediction in Pakistan, demonstrating that integrating environmental degradation indicators significantly improves traditional hydrological models. Among eight algorithms evaluated on a 5,500-sample dataset, Stacking Classifier emerged as the most effective, achieving 95.73% accuracy, 99.0% ROC-AUC, and 100% PR-AUC, while maintaining effective computational efficiency (3.4-minute training time, 1.2 GB memory usage), making it suitable for real-time deployment. Rainfall ($r = 0.78$), CO_2 emissions ($r = 0.79$), and deforestation ($r = 0.74$) were the most influential predictors, all statistically significant at $p < 0.001$. Regionally, Rawalpindi, Faisalabad, and Layyah showed the highest flood risk, while loam soils and agricultural land were the most vulnerable terrain types. The model's exceptional PR-AUC confirms its utility for early flood detection with low false-negative rates, supporting timely emergency responses. Policy insights derived from feature importance analysis suggest prioritizing reforestation, integrating emission reduction into flood planning, and enhancing water management protocols. Novel contributions include the first systematic integration of environmental indicators with hydrological data in the Pakistani context, a 5–8% performance improvement over existing models, a robust interpretability framework, and a scalable methodology adaptable to other regions. Robustness was confirmed through 94.3% temporal accuracy on 2023 holdout data and 93.7% ± 2.1% average district-level accuracy across Punjab. Statistical validation via McNemar's test ($\chi^2 = 247.3$, $p < 0.001$) affirms significant performance gains over traditional classifiers, establishing the model's reliability for practical flood risk management.

Future Enhancements

Future research should enhance dynamic flood forecasting by integrating IoT-based sensor networks, satellite data streams, and socio-economic indicators such as population density, infrastructure vulnerability, and economic exposure for more comprehensive risk assessments. Expanding the model's geographic coverage to provinces like Sindh, Khyber

Pakhtunkhwa, and Balochistan, alongside adopting advanced deep learning methods (e.g., transformer architectures and graph neural networks), will improve the modeling of complex spatial-temporal dependencies. Additionally, developing mobile applications with API-based real-time early warning systems will support community-level disaster preparedness and broaden the model's practical applicability across diverse regions.

Table 13: Recommended Future Developments

Priority Level	Enhancement	Expected Impact
High	Real-time data integration	+3-5% accuracy
High	Geographic expansion	National coverage
Medium	Mobile app development	Community adoption
Medium	Socioeconomic integration	Comprehensive risk assessment

i. STATEMENTS

Author Contributions

The authors verify their contribution to the paper in the following way. Conceptualization, Authors 1 and 2; methodology, Authors 1, and 3; software, Author 1; validation, Authors 1, 3, and 4; formal analysis, Author 1; investigation, Authors 1 and 5; resources, Authors 1 and 6; data curation, Author 1; writing – original draft preparation, Authors 1, and 3; writing – review All the authors revised the results and approved the final version of the manuscript.

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Data Availability Statement:

The data that support the findings of this study are available from the Corresponding Author, A.K.S., upon reasonable request.

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Ethics Approval

Not applicable.

Conflicts of Interest

The author(s) report no conflict of interest in terms of the present study.

Abbreviations

AI – Artificial Intelligence
ANN – Artificial Neural Network
CO_2 – Carbon Dioxide
EDA – Exploratory Data Analysis
FAO – Food and Agriculture Organization
GHG – Greenhouse Gas
GFW – Global Forest Watch
GLOF – Glacial Lake Outburst Flood
IPCC – Intergovernmental Panel on Climate Change
KNN – K-Nearest Neighbors
LGBM – Light Gradient Boosting Machine
LR – Logistic Regression
ML – Machine Learning
NDMA – National Disaster Management Authority
NOAA – National Oceanic and Atmospheric Administration
PCA – Principal Component Analysis
PMD – Pakistan Meteorological Department
RF – Random Forest
SDG – Sustainable Development Goal
SVM – Support Vector Machine
UNDP – United Nations Development Program
UNEP – United Nations Environment Program
USGS – United States Geological Survey
WRI – World Resources Institute
XGBoost – Extreme Gradient Boosting
NDVI – Normalized Difference Vegetation Index
LULC – Land Use Land Cover
GIS – Geographic Information System
RS – Remote Sensing
SMOTE – Synthetic Minority Over-sampling Technique
MAE – Mean Absolute Error
RMSE – Root Mean Square Error
R^2 – Coefficient of Determination

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